KARUNGA JUNIOR SCHOOL
GRADE 9 PRE-TECHNICAL STUDIES
END OF TERM 1 EXAMINATION

Teacher-prepared assessment resource | Kenya CBC

Learner's Name:	_____	Grade:	9
Date:	_____	Time:	2 hours 30 minutes

INSTRUCTIONS

1. Answer ALL questions.

2. Read each question carefully and show calculations where required.

3. Use labelled sketches where appropriate.

4. Follow workshop and electrical safety procedures at all times.

5. The examination carries 100 marks.

SECTION A: SHORT-ANSWER QUESTIONS (40 MARKS)

1. Design and Materials (8 marks)

(a) Define the term design brief. (2 marks)

(b) State two factors to consider when selecting materials for a product. (2 marks)

(c) Name two examples of natural materials used in making products. (2 marks)

(d) State two reasons for preparing a working drawing before production. (2 marks)

2. Workshop Tools and Safety (8 marks)

(a) Name two tools used for measuring or marking out materials. (2 marks)

(b) State two workshop safety rules. (2 marks)

(c) Give two examples of personal protective equipment used in a workshop. (2 marks)

(d) State two actions to take when a tool is damaged. (2 marks)

3. Materials and Processes (8 marks)

(a) State two properties of timber that make it useful in product making. (2 marks)

(b) Name two processes used to shape or form materials. (2 marks)

(c) What is meant by joining materials? (2 marks)

(d) Give two examples of joining methods. (2 marks)

4. Basic Electricity and Technology (8 marks)

(a) Define an electric circuit. (2 marks)

(b) Name two components of a simple electric circuit. (2 marks)

(c) State the function of a switch in a circuit. (2 marks)

(d) Give two electrical safety precautions. (2 marks)

5. Production, Finishing and Environment (8 marks)

(a) State two reasons for finishing a product. (2 marks)

(b) Name two finishing materials or methods. (2 marks)

(c) State two ways of reducing material wastage in a workshop. (2 marks)

(d) Give two examples of materials that can be reused or recycled in product making. (2 marks)

SECTION B: STRUCTURED QUESTIONS (60 MARKS)

6. Product Design Process (12 marks)

(a) Outline four stages a learner may follow when designing and making a simple useful product. (4 marks)

(b) A school needs a simple desk organiser made from locally available materials. State four requirements that should guide the design. (4 marks)

(c) Explain two reasons for evaluating a product after making it. (4 marks)

7. Workshop Practice and First Aid (12 marks)

(a) Explain three safe practices to observe when cutting or shaping materials in a workshop. (6 marks)

(b) State three actions to take when a learner sustains a minor cut in the workshop. (3 marks)

(c) Explain why tools should be returned to their designated storage places after use. (3 marks)

8. Materials, Joining and Finishing (12 marks)

(a) Differentiate between a temporary joint and a permanent joint. (4 marks)

(b) Describe two joining methods and give one suitable application for each. (4 marks)

(c) Explain two ways a suitable finish can improve a product. (4 marks)

9. Basic Electrical Systems (12 marks)

(a) Draw a simple circuit containing a cell, switch and lamp connected in series. Label the components. (4 marks)

(b) Explain what happens when the switch is opened and when it is closed. (4 marks)

(c) State two dangers of handling electrical appliances with wet hands or damaged cables. (4 marks)

10. Making and Evaluating a Product (12 marks)

(a) A learner makes a small phone stand from cardboard. State four checks that can be made to assess whether the product is fit for purpose. (4 marks)

(b) Explain three ways to minimise environmental harm during a school production activity. (6 marks)

(c) State two pieces of information that should be included in a simple product evaluation. (2 marks)

MARKING SCHEME
GRADE 9 PRE-TECHNICAL STUDIES END OF TERM 1

TOTAL: 100 MARKS

1. Design and Materials (8 marks)

(a) A short statement describing a problem, need or task that a product or design is expected to address. (2)

(b) Any two: strength, durability, cost, availability, safety, appearance, workability or environmental impact. (2)

(c) Any two: timber, cotton, leather, clay, natural fibres, bamboo or stone. (2)

(d) Any two: communicate dimensions and shape; guide accurate manufacture; show details; reduce errors and material wastage. (2)

2. Workshop Tools and Safety (8 marks)

(a) Any two: ruler, tape measure, try square, marking gauge, pencil, scriber. (2)

(b) Any two: follow instructions; wear PPE; keep floor clear; use correct tools; do not run or play; report hazards; secure workpieces. (2)

(c) Any two: safety goggles, gloves suitable for the task, apron/overalls, dust mask or safety footwear. (2)

(d) Stop using it; inform the teacher/supervisor; label or isolate it safely; do not attempt unauthorised repairs. Any two. (2)

3. Materials and Processes (8 marks)

(a) Any two: workable, strong, relatively light, renewable when responsibly sourced, insulating or attractive grain/appearance. (2)

(b) Any two: cutting, drilling, bending, moulding, casting, shaping, filing or sanding. (2)

(c) Connecting two or more parts/materials to form one assembly or product. (2)

(d) Any two: screws, nails, bolts and nuts, adhesive, welding, soldering, riveting or dovetail joint. (2)

4. Basic Electricity and Technology (8 marks)

(a) A complete path through which electric current can flow. (2)

(b) Any two: cell/battery, connecting wires, switch, lamp/bulb, resistor or buzzer. (2)

(c) It opens or closes the circuit to stop or allow current flow. (2)

(d) Any two: keep hands dry; do not touch bare wires; avoid overloaded sockets; report damaged cables; switch off before repairs; use correctly rated equipment. (2)

5. Production, Finishing and Environment (8 marks)

(a) Any two: improve appearance; protect from moisture/corrosion/wear; improve durability; make surfaces smoother/easier to clean. (2)

(b) Any two: paint, varnish, polish, oil, sanding, filing or applying a protective coating. (2)

(c) Any two: plan dimensions; measure accurately; reuse offcuts; maintain tools; cut efficiently; recycle suitable scraps. (2)

(d) Any two: cardboard, paper, timber offcuts, metal, plastics, fabric or glass, where safe and suitable. (2)

6. Product Design Process (12 marks)

(a) Any four stages: identify need/problem; research/investigate; develop ideas/sketch; select materials/tools; prepare working drawing; make prototype/product; test; evaluate and improve. (4)

(b) Any four: stable; holds stationery securely; safe edges; suitable size; durable; affordable; easy to make; uses available/recycled materials; attractive. (4)

(c) Any two explained: determine whether it meets the brief; identify faults; check safety/strength/usability; obtain user feedback; identify improvements. (4)

7. Workshop Practice and First Aid (12 marks)

(a) Any three explained: wear appropriate PPE; secure material in a vice/clamp; use the correct sharp tool; cut away from the body where appropriate; keep hands clear; maintain a stable posture; follow teacher instructions. (6)

(b) Stop work and inform the teacher; wash/clean the minor wound with clean water; apply suitable pressure with clean dressing; cover with sterile plaster/dressing; seek further help if bleeding is severe or wound is serious. Award any three safe actions. (3)

(c) Prevents accidents and loss; protects tools from damage; makes tools easy to find; keeps the workshop orderly. (3)

8. Materials, Joining and Finishing (12 marks)

(a) Temporary joints can be dismantled without significantly damaging components (e.g. screws/bolts); permanent joints are intended not to be separated without damaging the joint or components (e.g. welding/adhesive). (4)

(b) Any two described with application: screws for removable timber joints; nails for fixing timber; adhesive for paper/wood; welding for metal frames; riveting for sheet materials. (4)

(c) Any two explained: protects against moisture/corrosion; improves appearance; reduces wear; makes cleaning easier; improves smoothness and safety. (4)

9. Basic Electrical Systems (12 marks)

(a) Award 1 for each correctly shown and labelled cell, switch, lamp and connecting series path forming a complete circuit. (4)

(b) Open switch: circuit is broken and current does not flow, so lamp is off. Closed switch: circuit is complete and current flows, so lamp lights. (4)

(c) Any two explained: electric shock/electrocution; burns; short circuit; fire; damaged equipment. (4)

10. Making and Evaluating a Product (12 marks)

(a) Any four: stable on a flat surface; supports phone securely; suitable viewing angle; correct dimensions; smooth/safe edges; strong enough; easy to use; neat finish. (4)

(b) Any three explained: reduce/reuse/recycle materials; dispose of waste correctly; avoid harmful substances; use materials efficiently; conserve electricity/water; maintain good ventilation and dust control. (6)

(c) Any two: strengths and weaknesses; whether it meets design requirements; test results; user feedback; suggested improvements; materials and processes used. (2)

GENERAL MARKING GUIDANCE

Accept other technically correct answers and suitable local examples.

For sketches, award marks for correct components, labels, connections and clarity as specified.

For safety and first-aid questions, do not award marks for unsafe or unauthorised actions.

Teachers should ensure that workshop activities are supervised and that the assessed content has been taught.

This is an original teacher-prepared assessment resource and is not an official KICD examination.

END OF EXAMINATION